

Aryan Miriyala

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PROFESSIONAL SUMMARY

AI/ML systems-oriented computer science graduate student with experience in **Python** analysis code, **C++/OpenMP** runtime experimentation, workload benchmarking, reproducible modeling, and cloud-scale data pipelines. Built performance telemetry, adaptive scheduling, ML/RL experiments, and production data workflows with emphasis on quantitative analysis, technical communication, and systems tradeoffs.

EDUCATION

Bowling Green State University

Master of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH

Aug. 2024 – Aug. 2026

Bowling Green State University

Bachelor of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH

Jan. 2021 – Apr. 2024

EXPERIENCE

Artificial Intelligence & Computer Science Intern

SmartSolve Industries

May 2026 – Present

Bowling Green, OH

- Supported AI-enabled internal software planning, including a **QMS dependency register** for modeling relationships between quality documents and improving document traceability.
- Built secure internal software with **Next.js**, **TypeScript**, **PostgreSQL**, **Drizzle ORM**, **SSO**, and auth middleware for onboarding and org-chart workflows around sensitive employee data.
- Created a cross-platform **Docker** devcontainer for **WSL2/Colima**, standardizing reproducible development environments for AI-assisted software development and code review.

Data Engineering Intern

American Association of Insurance Services

May 2025 – Aug. 2025

Lisle, IL

- Automated a manual **SQL** billing workflow with **Python**, **PySpark**, and **AWS Glue**, processing **20+ TB** of production insurance data for **700+** member companies.
- Built data validation, reverse-ETL, and custom **AWS IAM** access patterns across **S3** and MDM workflows, improving controlled data access and reproducible production operations.

Software Engineering Intern

American Association of Insurance Services

May 2023 – May 2024

Lisle, IL

- Engineered **Python**/JSON backend automation generating **1,000+** production **SQL** tables across **Oracle/PostgreSQL**, supporting standardized data models across **10+** insurance lines.
- Built **Python/Pandas** workflows with multiprocessing, fuzzy matching, and hashing to clean large CSV datasets for production data-standardization pipelines.
- Automated database validation with **AWS Lambda** triggers that tokenized PII using **hashlib/boto3** and ingested structured JSON into **S3** for downstream workflows.

PROJECTS

Self-Adaptive Parallelism | *C++, OpenMP, Python, Docker, Linux*: Built a runtime research system with per-epoch telemetry and a UCB controller to tune scheduling policy, chunk size, and thread count across Mandelbrot, heat diffusion, and reduction workloads.

Performance Modeling Outputs | *Python, CSV telemetry, benchmarking, visualization*: Generated reproducible runtime, speedup, efficiency, Karp-Flatt, and scheduler-decision plots comparing adaptive scheduling against serial and fixed-policy baselines.

MarioRL Agent Comparison | *Python, PyTorch, Stable-Baselines3, Gym, TensorBoard*: Compared PPO and DQN agents with frame preprocessing, checkpoints, TensorBoard logs, generated clips, and reward/training-speed performance graphs.

Diff-Grounded PR Generation | *Python, OpenAI API, GitHub API, LLM evaluation*: Built an evidence-grounded generation pipeline using commits, diffs, linked issues, and metadata to produce structured PR descriptions and evaluate source-grounded outputs.

HealthTrend Data Pipeline | *Spark, PySpark, Kafka, Hadoop HDFS, Docker Compose*: Built a containerized pipeline routing CSV patient data and JSON health events into distributed storage and stream-processing workflows.

TECHNICAL SKILLS

Systems & Performance: C++, OpenMP, Python, Linux, Docker, Make, runtime instrumentation, adaptive scheduling, UCB optimization, workload characterization, throughput/latency/efficiency analysis, CSV telemetry, data visualization

AI/ML & Modeling: PyTorch, Stable-Baselines3, PPO, DQN, Gym, TensorBoard, reinforcement learning, LLM inference, RAG, prompt engineering

Data & Analysis: Pandas, PySpark, Spark, SQL, AWS Glue, S3, Lambda, IAM, Kafka, Hadoop HDFS, data validation, reproducible pipelines

Software Engineering: Git, Docker Compose, TypeScript/JavaScript, Next.js, Node.js, Express, PostgreSQL, Oracle, REST APIs, technical documentation